Quantum Measurement, Minkowski Microstates, and Matter Channels

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Abstract

We show how quantum state reduction naturally plays the role of a microstate in thermodynamic approaches to gravity and develop an information-theoretic language to explore the connection with a concrete construction. Eliminating temperature between Landauer's principle and the Unruh effect yields a purely kinematic acceleration-dependent bound on the energy cost of information transfer. Motivated by this, we construct a helicity-anticorrelated Bell state, emitted from a common causal past, that drives the Unruh response in both Rindler wedges of flat Minkowski space. We briefly discuss how proper time can be measured as a property of matter, and then connect the matter action, event by event, to a Gibbons–Hawking–York boundary term and a horizon area increment. Using a pp-wave cut-and-paste surgery, we show how space is dynamically advanced along the horizons. This newly generated geometry carries a concrete label tied to the entangled helicity, which is subsequently observed across the causal horizons, allowing for the measurement of space simultaneously in geometric and informational terms.

1 Introduction

Jacobson derives the Einstein equations from entropy and temperature on local Rindler horizons [1]; Verlinde derives gravitational forces from entropy gradients on holographic screens [2]. Both frameworks operate at the level of bulk thermodynamics and leave open a prior question: *what physical process underlies each individual entropy increment?* We will show how the answer can be quantum measurement by developing an information-theoretic language that connects "thrust" microstates, that can drive the Unruh effect [3], to current literature.

Before we dig in though, we first briefly outline our high level argument about how quantum measurement naturally connects to gravity. The general setup is pictured in the commutative[1] diagram in Figure 1, which we will walk through from the top right clockwise around the bottom to the top left. Quantum state reduction always happens due to an observation. Any observation is mediated by a physical interaction: information transfer necessarily involves an exchange of energy and momentum between the observed system and the detector, a discrete impulse. And then the equivalence principle implies that this impulse is locally indistinguishable from the effect of a gravitational field.

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[1] This refactors the Diósi–Penrose postulate in the categorical sense: the dashed arrow is not assumed but derived as the composite of the three solid arrows. Diósi–Penrose postulates a direct link from gravitational effects to state reduction; the present diagram factors that link through acceleration and observation, grounding each step independently.

$$
\begin{tikzcd}
\text{Local Gravitational Effects} \arrow[r, dashed, "\text{Diósi–Penrose}"] & \text{Quantum State Reduction} \\
\text{Acceleration} \arrow[u, "\text{local equivalence \S5}"'] \arrow[r, "\text{energy–momentum transfer \S2}"'] \arrow[ur, phantom, "\circlearrowright\ \S4"] & \text{Observation} \arrow[u, "\text{Bayesian update \S3,\S6}"']
\end{tikzcd}
$$

Figure 1: Solid arrows denote subjective grounding relations for a given observer: $A \to B$ means every instance of B is physically realized by an instance of A. This paper explores the solid arrows. The dashed arrow is the Diósi–Penrose postulate.

1.1 Roadmap

The paper develops each solid arrow of Figure 1 in turn, with Section 4 providing the information-theoretic framework that unifies them into a single physical unit, a communication between matter channels mediated by a massless quantum.

The **energy–momentum transfer** arrow is developed in Section 2: substituting the Unruh temperature into Landauer's principle eliminates temperature entirely, yielding a purely kinematic bound on the energy cost of information acquisition under acceleration. The thrust quantum, the photon carrying the impulse, is the physical carrier of this bound.

The **Bayesian update** arrow is developed in Sections 3 and 6: the POVM-to-PVM transition selects a single non-thermal microstate from the Minkowski vacuum, the helicity outcome is recorded as a spatial bit in matter-memory, and the anti-correlation between right and left Rindler wedges is enforced by their common causal ancestry in the past wedge.

The **local equivalence** arrow is developed in Section 5: the absorbed photon sources an Aichelburg–Sexl pp-wave at the Rindler horizon, the Raychaudhuri equation gives a discrete area increment $\Delta A = 8\pi G\hbar\omega$, and the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ closes the consistency loop between geometry and information.

Section 4 is the conceptual core. It explains why these three arrows are aspects of one physical unit: a communication event that transfers one quantum of action between matter-memories. Several of the thermodynamic identities recovered along the way, the area law, the Bekenstein bound, the Unruh spectrum, are well established; what is new is locating them microstate by microstate in a single physical event, rather than recovering them only as a bulk average.

2 The Unruh–Landauer Bound and Motivating Heuristics

A structuring choice runs throughout the paper. We work in the non-thermal global Minkowski description rather than the thermal Rindler description: thermality is a derived consequence of restricting to one wedge [4, 5], not a primitive, and individual microstates are non-thermal.

This section has two distinct parts. The Unruh–Landauer relation of Section 2.1 is an exact result that the construction layer inherits directly. substituting the Unruh temperature into Landauer's bound eliminates temperature entirely, leaving a purely kinematic inequality depending only on acceleration. The subsequent derivations are explicitly Verlinde-style heuristics [2]: the same actors appear, acceleration, the Unruh relation, Newton's laws, but thermality is eliminated rather than embraced. Where Verlinde deploys the thermal Rindler frame as the mechanism connecting entropy to force, the Unruh–Landauer relation eliminates temperature entirely, and Newton's laws then recover standard black hole thermodynamics directly from the resulting kinematic bound. Thermality is a consequence of restricted causal access to one Rindler wedge [4, 5], not a primitive.

2.1 The Unruh–Landauer Relation

Landauer's principle [6] establishes $k_B T$ as the natural energy scale of one distinguishable degree of freedom in a thermal environment at temperature T: any physical carrier representing ΔH nats must have at least this energy to be distinguishable from thermal noise,

$$\Delta E \geq k_B T \, \Delta H. \tag{1}$$

For an observer undergoing proper acceleration a, the Unruh effect associates a temperature

$$T = \frac{\hbar a}{2\pi k_B c} \tag{2}$$

with the local vacuum. Substituting into Landauer's bound eliminates $k_B T$, yielding

$$\boxed{\Delta E \geq \frac{\hbar a}{2\pi c} \, \Delta H.} \tag{3}$$

Temperature has dropped out entirely: equation (3) is a temperature-independent lower bound on the energy of the physical carrier required to register one nat of information under proper acceleration a; acceleration sets the minimum energy scale of the carrier. The full thermodynamic accounting, where the Landauer erasure cost appears, is deferred to future work.

The connection between Landauer's principle and black hole thermodynamics has been noted previously. Cortês and Liddle [7] showed that Hawking evaporation saturates the Landauer bound at the Hawking temperature, identifying the two results as expressions of the same process at the level of bulk evaporation. Related bounds on the energy cost of information acquisition in curved spacetime have been derived using the second law of thermodynamics applied to relativistic communication channels [8]. The present bound (3) differs in two respects: temperature is eliminated entirely by substituting the Unruh relation, yielding a purely kinematic bound depending only on acceleration, and the bound is applied to individual carrier quanta rather than to bulk evaporation.

This Unruh–Landauer bound is saturated for a photon at one bit when the wavelength equals the horizon distance up to a factor of $4\pi^2/\ln 2$:

$$\lambda_{\text{sat}} = \frac{4\pi^2}{\ln 2} \cdot \frac{c^2}{a} \sim d_{\mathcal{H}}, \tag{4}$$

where $d_{\mathcal{H}} = c^2/a$ is the proper distance to the Rindler horizon. The saturating photon is the fundamental mode of the causal diamond defined by the observer and the horizon, with wavelength commensurate with the horizon distance $d_{\mathcal{H}}$, too short to be super-horizon and too long to deposit more than one bit. This identifies the horizon distance as the natural length scale of a single-bit measurement event, and the Rindler horizon as the surface at which the Landauer bound is geometrically realized.

For realistic accelerations the saturation wavelength is enormous, at Earth surface gravity $a = 9.8\ \text{m}\,\text{s}^{-2}$, $\lambda_{\text{sat}} \approx 55$ light years, so the single-bit bound is practically always satisfied for terrestrial observers, and is only approached near astrophysical or Planck-scale accelerations.

2.2 Momentum Transfer and Acceleration

Note on method. Sections 2.2–2.5 are Verlinde-style dimensional analysis. They use Newtonian gravity and recover correct results at the Schwarzschild scale by the same coincidence exploited by Verlinde [2] and Bekenstein [9]: the Newtonian inverse-square law with a point mass reproduces the Schwarzschild geometry at the horizon scale, even though it formally breaks down there. This coincidence is specific to the spherically symmetric case; other matter distributions, including the bilocal source of Section 3, or the full Einstein equations, will in general give different results. The construction layer of Sections 3–5 makes no use of these heuristics. We make no attempt to disguise any of this.

In what follows, standard results are *reproduced as heuristic consequences* of the Unruh–Landauer relation and Newton's laws. The energy transfer ΔE implied by the relation is carried by a physical quantum, matter or radiation, mediating the measurement interaction as a *thrust quantum.* For a relativistic carrier, the associated momentum transfer is

$$\Delta p = \frac{\Delta E}{c}. \tag{5}$$

A measurement event localized to a spatial scale r is bounded by a causal diamond with temporal extent $\Delta t = 2r/c$. This identification is the heuristic input from which the subsequent results follow, then

$$a = \frac{\Delta p}{m_d \Delta t} = \frac{\Delta E}{2m_d r}. \tag{6}$$

This is closely related to Verlinde's entropic gravity [2]: Verlinde identifies holographic thrust on information screens with the origin of inertia itself, and Newton's laws are a common ingredient in both constructions. The present approach differs in grounding each entropy increment in a discrete measurement event, the measurement impulse playing the role of Verlinde's holographic thrust, rather than in a coarse-grained thermodynamic gradient.

2.3 The Schwarzschild Radius

We match the kinematic acceleration of the detector to the gravitational field of the thrust quantum using Newton's laws. By $E = mc^2$, the energy quantum ΔE gravitates with effective mass $\Delta E/c^2$, producing a Newtonian gravitational acceleration

$$a = \frac{G\Delta E}{c^2 r^2} \tag{7}$$

at distance r from the detector. Equating this with the measurement-induced acceleration from equation (6),

$$\frac{\Delta E}{2m_d r} = \frac{G\Delta E}{c^2 r^2}. \tag{8}$$

The energy ΔE of the thrust quantum cancels entirely: the Schwarzschild radius is independent of the energy of the carrier. Solving for r:

$$r = \frac{2Gm_d}{c^2} \equiv r_s, \tag{9}$$

which is precisely the Schwarzschild radius of the detector mass m_d. The cancellation of ΔE is the physical content of the result: whatever energy the thrust quantum carries, the self-consistent horizon scale is set by the detector alone.

2.4 Surface Gravity

Evaluating the acceleration at $r = r_s$ gives the surface gravity of the detector:

$$\kappa = \frac{Gm_d}{r_s^2} = \frac{c^4}{4Gm_d}, \tag{10}$$

in agreement with the standard Schwarzschild result, as expected from the Verlinde coincidence. This reproduces the acceleration appearing in the Unruh temperature $T = \hbar\kappa/2\pi k_B c$, providing an internal consistency check on the heuristics.

2.5 Bekenstein–Hawking Entropy

Returning to the Unruh-Landauer relation (3) and substituting the surface gravity κ for a, justified by the equivalence principle: the local quantum field theory at the black hole horizon is the same as in the Rindler frame [10, 11], the entropy increment associated with an energy increment $\Delta E = c^2 \Delta m_d$ is

$$\Delta H \leq \frac{2\pi c}{\hbar\kappa}\Delta E = \frac{8\pi G m_d}{\hbar c^3} \cdot \Delta m_d c^2. \tag{11}$$

Expressing this in terms of the change in Schwarzschild area $\Delta A = 32\pi G^2 m_d \Delta m_d / c^4$:

$$\Delta H \leq \frac{\Delta A}{4\ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \tag{12}$$

which is the Bekenstein-Hawking area law. Each measurement event contributes a discrete entropy increment proportional to the area imprinted on the horizon by the associated energy-momentum transfer.

This derivation parallels Jacob Bekenstein's original argument [9], in which the entropy increase of a black hole is inferred from the minimal energy required to localize and drop a system across the horizon; here, the same structure emerges from the minimal energy cost of registering information at the horizon scale.

The two bounds are saturated simultaneously when $a = c^2/R$, which is exactly the condition for being *at a horizon*. The horizon is therefore not just where thermodynamics happens to work out, it is the unique locus where the minimum cost of information acquisition equals the maximum information content of the region.

Substituting the horizon condition $a = c^2/R$ directly into the Unruh–Landauer bound (3) yields

$$\Delta H \leq \frac{2\pi R\,\Delta E}{\hbar c}, \tag{13}$$

which is precisely the Bekenstein universal entropy bound [12] on the entropy-to-energy ratio for a system of size R. The Unruh–Landauer relation and the Bekenstein bound are therefore the same inequality: the former is its acceleration-frame expression, the latter its position-space expression, and the Rindler horizon is the surface at which the two descriptions coincide. The covariant generalisation of the Bekenstein bound to arbitrary null hypersurfaces is the Bousso bound [13], which bounds entropy on any non-expanding null congruence by the area of its generating surface. The Raychaudhuri derivation of Section 5 operates precisely on such a congruence; the present construction can be read as a microstate-level realisation of the Bousso bound, with each communication contributing one Planck-area patch to the light-sheet.

Pesci [14] derived the covariant entropy bound from the Unruh temperature by applying the second law of thermodynamics across local Rindler horizons, the closest precursor to the present argument. The present bound equation (3) differs in two respects: temperature is eliminated entirely by substituting the Unruh relation into Landauer's principle rather than retained as an input to the second law, yielding a purely kinematic bound depending only on acceleration; and the bound is applied to individual carrier quanta rather than to a bulk thermodynamic flux. The identification of equation (3) with the Bekenstein bound [12] at the horizon condition $a = c^2/R$, and the reading of both as the same inequality in different frames, appears to be new in this form.

3 Bilocal Microstate Construction

From now on we use natural units $\hbar = c = 1$ unless otherwise specified; factors of $\hbar$ and c are restored where they aid physical interpretation.

The arguments of Section 2 motivate a heuristic relationship between information acquisition, acceleration, and gravitational effects, but leave open the question of what a single microstate looks like concretely. In standard treatments, the Unruh effect is derived by coupling a detector to a pre-existing acceleration and reading off a thermal response from the vacuum. In [3] this logic was inverted for a scalar field: a source placed in the past injects non-thermal entangled excitations into the field, and the observer's accelerated response was shown to be driveable by these source-induced excitations. We extend that construction here to the spin-1 electromagnetic field, upgrading from scalar to helicity-carrying Bell states and making the single-bit structure of each measurement event explicit.

3.1 A Source and Two Observers

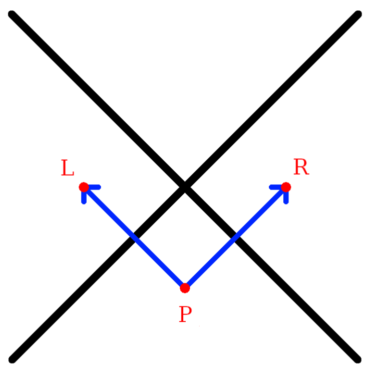

Figure 2: A source in the past, $\mathcal{P}$, emits an entangled photon pair that accelerates $\mathcal{R}$ into the right Rindler wedge and $\mathcal{L}$ into the left Rindler wedge. The two null directions of propagation define the null coordinates $u = t - z + 1$ and $v = t + z + 1$, with the right-moving photon propagating along $u = 0$ and the left-moving photon along $v = 0$; these are the z^- and z^+ axes of the double-null coordinate system of Section 3.4.

Three massive bodies are placed at spacetime positions $(t, z) = (0, 1)$, $(0, -1)$, and $(-1, 0)$, named $\mathcal{R}$ – *Right*, $\mathcal{L}$ – *Left*, and $\mathcal{P}$ – *Past* respectively. $\mathcal{P}$ serves as the common causal ancestor of the accelerations of $\mathcal{R}$ and $\mathcal{L}$ (see Figure 2). $\mathcal{P}$ emits an entangled pair of spin-1 massless photons with momenta $\pm k$ and Rindler frequency $\omega = \omega_k$. The source exhibits no net back-reaction in either linear or angular momentum[2]. The two photons propagate into the left and right Rindler wedges, where $\mathcal{L}$ and $\mathcal{R}$ detect them

[2]The momenta are perfectly balanced by construction: the two photons carry spatial momenta $+k\hat{z}$ and $-k\hat{z}$ respectively, so the total linear momentum vanishes, $k + (-k) = 0$. The angular momentum vanishes term by term: in each helicity assignment $(\lambda_R, \lambda_L) = (+1, -1)$ and $(-1, +1)$, the total helicity is $\lambda_R + \lambda_L = (+1) + (-1) = 0$ and $(-1) + (+1) = 0$ respectively, and the symmetric Bell state superposition (26) preserves this since both terms contribute zero independently

respectively via absorption, the momentum impulse of which accelerates each detector into its respective Rindler wedge; each observer thereby registers a helicity eigenvalue and acquires one classical bit of information. The emission event at $\mathcal{P}$ naturally defines null coordinates $u = t - z + 1$ and $v = t + z + 1$, centered so that $\mathcal{P}$ lies at $u = v = 0$, the right-moving photon propagates along $u = 0$, and the left-moving photon propagates along $v = 0$. The three-party structure $\mathcal{P}$–$\mathcal{R}$–$\mathcal{L}$ realises a form of tripartite entanglement[3] with a natural interior reconstruction interpretation [15, 16].

3.2 PVM Localization and the Elimination of Coarse-Graining

The Unruh mode plays a double role in this construction. As a QFT object it is a coefficient in a mode expansion, a solution to the wave equation analytically continued from the Rindler wedge into all of Minkowski space. But once $\mathcal{P}$ is specified as a real physical source via the squeezing interaction (equation(20) upcoming), the mode function acquires a second interpretation: it is a probability amplitude in the Born rule sense, encoding where $\mathcal{R}$ is likely to detect the emitted quantum. This is the inversion of the standard Unruh–DeWitt logic. Rather than coupling a detector to a pre-existing acceleration and reading off a thermal distribution by averaging over all Rindler frequencies, we front-load the logic at $\mathcal{P}$: the source fixes the frequency ω_k and the mode function then specifies the probability amplitude for detection.

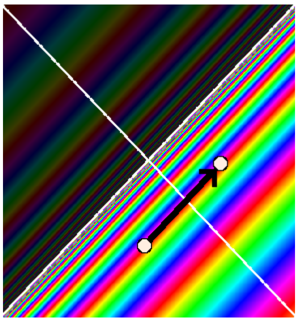

Figure 3: The points $\mathcal{P}$ and $\mathcal{R}$ are marked with white dots, connected by the null geodesic (black arrow). The rainbow coloring shows the Unruh mode phase structure across Minkowski space. A PVM projection onto this null line selects a single frequency from the thermal superposition, replacing the full delocalized mode with the localized non-thermal microstate of equation (26).

Longitudinal localization. The primary localization is in the longitudinal (t, z) plane, which is the setting of Figure 3 and of the parabolic cylinder function analysis of [3]. In that work, a Minkowski mode φ_q^* multiplied pointwise by a Gaussian shaping function $N_{\mu,\sigma}$ of width σ in (t, z), then paired via the Klein–Gordon inner product with a Rindler mode r_k, yields an overlap expressible in terms of parabolic cylinder functions $D_\nu(-\zeta)$. The Gaussian could equivalently have been applied to the Rindler side; what matters is that σ controls the degree of longitudinal localization of the source-detector pair. The Stokes phenomenon in the asymptotics of $D_\nu(-\zeta)$ as σ varies drives a transition between two physically distinct regimes: $\sigma \to \infty$ gives the thermal regime where the $e^{-\frac{1}{4}\zeta^2}$ branch dominates, and $\sigma \to 0$ gives the localized non-thermal regime where

[3]Neither $\mathcal{R}$ nor $\mathcal{L}$ alone can distinguish a correlated Bell-state emission at $\mathcal{P}$ from an uncorrelated thermal source: each observer's reduced density matrix is thermal, exactly as in the standard Unruh effect. Together, their two anticorrelated bits reveal the correlation structure of the emission event, identifying it as a specific Bell state at a specific frequency, and providing a microstate-level analogue of black hole interior reconstruction, where the interior is recovered as classical correlations in the joint record of complementary exterior observers.

the $e^{+\frac{1}{4}\zeta^2}$ branch dominates, with the Stokes line at $\arg\zeta = \pi/4$ marking the boundary between them.

In the present paper the same Gaussian shaping function is reread as a Kraus operator acting on the quantum state.

$$K_\sigma = \int e^{-u^2/2\sigma^2}\, |u\rangle\langle u|\, du, \tag{14}$$

where $u = t - z + 1$ is the null coordinate[4] defined in Section 5. The operator K_σ together with its complement

$$K_\sigma^\perp = \int \sqrt{1 - e^{-u^2/\sigma^2}}\, |u\rangle\langle u|\, du, \tag{15}$$

satisfying $K_\sigma^\dagger K_\sigma + (K_\sigma^\perp)^\dagger K_\sigma^\perp = \mathbf{1}$ by construction, defines a one-parameter POVM family; the Stokes transition between its two asymptotic regimes is illustrated in Figure 4.

The Stokes transition of [3] is therefore a field-theoretic regularization result and the analytic signature of the POVM-to-PVM transition: the same mathematical object is being read in two complementary languages, field theory and measurement theory, with the Stokes line marking the boundary between them. Specifically, the result imported from [3] is that the overlap between a Gaussian-shaped Minkowski source of width σ and a Rindler mode at frequency ω is expressible in terms of parabolic cylinder functions $D_\nu(-\zeta)$, whose asymptotic behaviour undergoes a Stokes transition at $\arg\zeta = \pi/4$. In the present paper this result is reread as the operator statement that the Kraus family (14) interpolates between a POVM ($\sigma \to \infty$) and a PVM ($\sigma \to 0$), with the Stokes line marking the boundary between the two regimes.

The helicity label λ enters the longitudinal mode function as a spectator index and does not affect the u-dependence of the Kraus operator (14); the Stokes transition analysis of [3], which is carried out in 1+1 dimensions, therefore applies independently to each helicity sector without modification. The extension to 3+1 dimensions introduces the transverse coordinates (x, y) as an independent sector: because the wave equation in double-null coordinates separates, the longitudinal Stokes analysis is unaffected by the transverse directions, which are handled separately by the transverse Kraus family (27).

Thermality is not a primitive property of the vacuum but a consequence of epistemic coarse-graining: the global Minkowski state $|0\rangle_M$ is pure and fixed throughout, and what changes as the source-detector pair $(\mathcal{P}, \mathcal{R})$ is specified is the resolution of the observer's ignorance about which null geodesic carried the quantum. This distinction is operationally sharp [18, 19]: *mode shaping* is a preparation that modifies the physical field; *PVM projection* is a measurement that conditions the observer's description on causal data without touching the ontic state. The Gaussian width σ of the Kraus family (14) admits both readings simultaneously, as a source localization parameter in the preparation sense of [3], and as an epistemic resolution parameter in the measurement sense here, with the Stokes line marking the boundary at which the two readings give qualitatively different physics. Specifying $\mathcal{R} = (0, 1)$ and $\mathcal{P} = (-1, 0)$ is therefore the epistemic $\sigma \to 0$ limit: the Kraus operator sharpens to a PVM projector onto the unique null geodesic connecting them, selecting a definite frequency ω_k and resolving both coarse-grainings responsible for thermality. The direction of explanation is thereby reversed relative to standard holographic arguments: the emission event at $\mathcal{P}$ is primary, and the geometry, the AS pp-wave, the horizon area increment, the BMS supertranslation, is the ontic output of the subsequent measurement events, while the assignment of that output to a particular observer's microstate record remains epistemic.

Transverse localization. The transverse (x, y) directions require a separate and independent localization. A Rindler plane wave at frequency ω is a coherent superposition

[4] We use ζ for the Stokes variable of [3] to avoid confusion with the longitudinal spatial coordinate z.

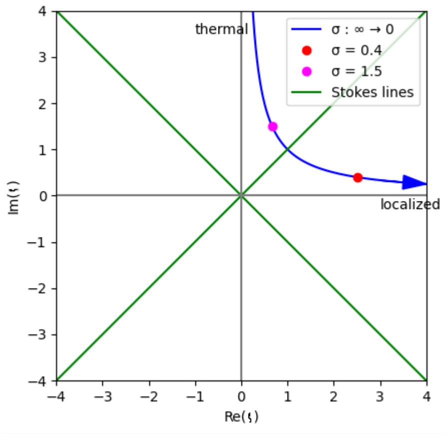

Figure 4: Trajectory of $\zeta = iq\sigma + \mu/\sigma$ in the complex plane as σ interpolates between the two limits of the Kraus family (14). The thermal limit $\sigma \to \infty$ corresponds to ζ on the positive imaginary axis; the localized PVM limit $\sigma \to 0$ corresponds to ζ on the positive real axis. The green Stokes lines at $\arg\zeta = \pi/4$ mark the boundary between the two asymptotic regimes of the parabolic cylinder function $D_\nu(-\zeta)$, and therefore the boundary between the POVM and PVM descriptions of the longitudinal localization. Figure from [3].

over all possible source-detector pairs $(\mathcal{P}', \mathcal{R}')$ connected by null geodesics at that frequency, uniform in the transverse plane. The longitudinal and transverse localizations decouple because the wave equation in double-null coordinates separates: the u-dependence and the (x, y)-dependence of the mode function enter independently, so the two Kraus projections can be applied in either order without interference. Specifying $\mathcal{P} = (-1, 0)$ and $\mathcal{R} = (0, 1)$ performs a second PVM projection, now in the transverse directions, selecting the unique null geodesic and resolving the remaining coarse-graining over transverse position. The transverse shaping is controlled by an independent width $\sigma_\perp$, which we treat as a free parameter in Section 3.7 and whose physical consequences are developed in Section 5.

Since the two widths σ and $\sigma_\perp$ control physically distinct aspects of the construction they need not be sent to zero together. The longitudinal σ governs the POVM-to-PVM transition and the thermal-to-non-thermal transition analyzed via the Stokes phenomenon; Section 3 works entirely in the $\sigma \to 0$ PVM limit. The transverse $\sigma_\perp$ controls the spatial profile of the horizon perturbation and the BMS mode content: $\sigma_\perp \to \infty$ produces a planar impulsive wave exciting a single BMS supertranslation mode, while $\sigma_\perp \to 0$ recovers the Aichelburg–Sexl pp-wave with logarithmic shift $\Delta v \propto -\ln r^2$ exciting the full BMS spectrum. The area increment $\Delta A = 8\pi G\hbar\omega$ and the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ in Section 5, which serve as the geometric consistency checks of the framework, are independent of $\sigma_\perp$, fixed by energy conservation alone.

The microstate. The result of both projections is an individually non-thermal microstate carrying a definite Rindler frequency ω and a definite longitudinal position on the null geodesic from $\mathcal{P}$ to $\mathcal{R}$. Thermality is not a property of any single microstate but of the ensemble; it emerges when one models the detector response within a wedge as a sum over all Rindler modes, as in the standard Unruh–DeWitt treatment [10, 21], and is a consequence of that mode-averaging rather than a premise of the construction. The individual non-thermal microstate is what the PVM projection onto the $\mathcal{P}$–$\mathcal{R}$ null geodesic selects; the thermal ensemble is what remains when neither coarse-graining is resolved and $\mathcal{L}$ is traced out.

A note on the $\sigma \to 0$ idealization: throughout this paper the PVM limit is understood as the idealization of a physical construction that lives at small but finite σ. At exactly $\sigma = 0$ the Kraus operator (14) becomes a projector onto a set of measure zero in the continuum $|u\rangle$ basis and is no longer normalizable in the strict sense; similarly, the AS longitudinal shift $\Delta v \propto -\ln r^2$ diverges on axis at $r = 0$ in the strict localized limit. Both divergences are regularized by keeping $\sigma > 0$. The parameter σ therefore plays a unified role: it is simultaneously the width of the Kraus operator in the longitudinal direction, the transverse profile width of the photon mode, and the regularization scale of the AS singularity, all controlled by the same physical quantity, the photon wavelength $\sigma \sim c/\omega$. All sharp statements about individual microstates should be understood as holding in the limit $\sigma \to 0^+$ rather than at exactly $\sigma = 0$.

3.3 Helicity Structure and PCT Symmetry

We now upgrade this construction from a spin-0 scalar field in [3] to the spin-1 electromagnetic field with a superposition of helicity pairs in a Bell state, introducing one unknown binary degree of freedom, the helicity outcome, undetermined until measured, which is the single bit responsible for the remaining thermality in the ensemble. The photon carries one quantum of action $\hbar$ and energy $E = \hbar\omega$, identifying the thrust quantum of Section 2 with a single photon whose polarization outcome, upon detection, is the classical bit acquired by each observer. The Bell state is the natural completion of the PCT symmetry between wedges, which by the Bisognano–Wichmann theorem [5] is implemented by the modular conjugation operator J mapping the right wedge algebra to the left wedge algebra, and as we discuss in Section 5, the photons imprint a definite geometric signature onto the horizon.

The anticorrelated helicity assignment traces to the PCT symmetry implemented by the Bisognano–Wichmann modular conjugation J, which maps the right-wedge algebra to the left-wedge algebra and simultaneously flips $\lambda \to -\lambda$; the Bell state (26) is the natural state in this momentum sector invariant under that operation. The common algebraic origin of this anticorrelation in the Lorentz algebra decomposition $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$, developed in Section 9.

3.4 Double-Null Gauge

The emission event naturally defines two null directions: the right-moving photon has wave-vector $k_R^\mu = (\omega, 0, 0, k)$ and the left-moving photon has wave-vector $k_L^\mu = (\omega, 0, 0, -k)$, with equal and opposite spatial momenta. These two null directions are precisely the z^+ and z^- axes of double-null coordinates

$$z^\pm = \frac{1}{\sqrt{2}}(t \pm z) = \frac{1}{\sqrt{2}}(u - 1 \pm (v - 1)), \tag{16}$$

with transverse coordinates $\mathbf{z}_\perp = (x, y)$. Note that $z^\pm$ are centered on the Minkowski origin $(t, z) = (0, 0)$ rather than on $\mathcal{P}$; the coordinates u, v defined in Section 3 are the $\mathcal{P}$-centered versions and are used throughout the remainder of the paper. Working in this frame we impose the symmetric double-null gauge condition

$$A_+(z) = 0, \qquad A_-(z) = 0, \tag{17}$$

so that the gauge field is supported entirely on the transverse components $\mathbf{A}_\perp = (A^x, A^y)$. This gauge respects the PCT symmetry that swaps the two wedges ($z^+ \leftrightarrow z^-$) simultaneously with ($R \leftrightarrow L$), and therefore treats both observers symmetrically by construction. The residual gauge freedom is fixed by requiring $\partial_i A^i = 0$ in the transverse plane, leaving precisely the two physical helicity degrees of freedom $\epsilon^i_{(\lambda)}$ with $\lambda = \pm 1$.

The lack of manifest four-dimensional covariance in equation (17) is a consequence of aligning the gauge with the pair of physical null directions selected by the source, rather than an arbitrary choice. The full Lorentz group is broken to the little group preserving the pair $\{k^\mu_R, k^\mu_L\}$, which is the two-dimensional Euclidean group $ISO(2)$ acting on the transverse plane. This is precisely the little group of massless particles, and the two helicity eigenstates $\lambda = \pm 1$ are its irreducible representations. All physical results are independent of the gauge choice: the helicity eigenvalues $\lambda = \pm 1$ are representations of the massless little group $ISO(2)$ and are therefore Poincaré-covariant objects independent of how the gauge is fixed.

3.5 The Bilocal Source

In double-null gauge the electromagnetic field is characterised by its transverse components alone. The bilocal driving source is symmetrized over both helicity assignments,

$$J^{ij}(x,y) = g\left(\epsilon^{i*}_{(-1)}\,\epsilon^{j}_{(+1)} + \epsilon^{i*}_{(+1)}\,\epsilon^{j}_{(-1)}\right) f_L(x)\, f_R(y), \tag{18}$$

where f_L and f_R are normalised mode profiles supported on the left and right wedges respectively, and $\epsilon^i_{(\lambda)}$ are the circular polarization vectors

$$\epsilon^i_{(+1)} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \qquad \epsilon^i_{(-1)} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}. \tag{19}$$

The two terms in (18) are related by the PCT transformation in the sense of Bisognano–Wichmann [5], which maps the right wedge algebra to the left wedge algebra and simultaneously flips $\lambda \to -\lambda$.

More precisely, the modular conjugation J of Bisognano–Wichmann maps a right-wedge creation operator $a^{R\dagger}_{k,(\lambda)}$ to a left-wedge creation operator $a^{L\dagger}_{-k,(-\lambda)}$, implementing PCT between wedges. The Bell state (26) is a natural two-photon state at fixed frequency ω and fixed momentum pair $(k, -k)$: within this momentum sector, the anticorrelated helicity assignment and the symmetric superposition respect the J-symmetry between wedges, since J flips both helicity and wedge simultaneously. The bilocal source (18) is therefore a natural source, consistent with the algebraic symmetry of the Rindler vacuum, and the simplest one we could construct with the required properties.

The symmetrization is consistent with PCT symmetry and angular momentum conservation at the emission vertex. Other source geometries are possible; this one is chosen for its simplicity and the transparency of its connection to the algebraic structure of the Rindler vacuum. A scalar source carries zero angular momentum, so the emitted pair must satisfy $\lambda_R + \lambda_L = 0$, which forces the two helicities to be anticorrelated: the only possibilities are $(\lambda_R, \lambda_L) = (+1, -1)$ or $(-1, +1)$. The Bisognano–Wichmann modular conjugation J maps the first assignment to the second and vice versa, and the symmetric superposition of the two is the simplest state respecting this exchange symmetry. The total angular momentum vanishes in both terms independently.

The bilocal source selects a specific microstate from the Minkowski vacuum by a PVM projection onto the $n = 1$ sector at fixed ω and fixed helicity pair $(\lambda, -\lambda)$, in exact analogy with the scalar construction of [3]. No perturbative expansion in a coupling constant is required: the helicity-anticorrelated photon pair is already present in the vacuum decomposition (24), and the bilocal source selects it via the PVM projection of Section 3.2. The operator content of this selection is

$$a^{L\dagger}_{-k,(-1)}\, a^{R\dagger}_{k,(+1)} \;+\; a^{L\dagger}_{-k,(+1)}\, a^{R\dagger}_{k,(-1)} \;+\; \text{h.c.}, \tag{20}$$

where the two creation operator pairs correspond to the two PCT-related helicity assignments, each carrying total angular momentum zero, consistent with emission from a scalar source at $\mathcal{P}$. This is the vector-field analogue of the scalar squeezing interaction [22], with the PCT symmetry between wedges made explicit; the connection to the quantum optics squeezing literature is structural rather than perturbative.

Each photon carries energy $E = \hbar\omega$ and one quantum of action $\hbar$, identifying the thrust quantum of Section 2 with a concrete physical object: the minimum actionable event is precisely a single photon. Its polarization, undetermined until measured, is the quantum bit whose outcome constitutes the classical record acquired by each observer upon detection.

The Unruh–Landauer bound is saturated not at the level of an individual photon but at the level of the source rate. The acceleration a does not fix the frequency ω of each selected quantum; rather, it determines the rate $\dot{N}$ at which the bilocal source selects entangled pairs in order to implement that acceleration, with the total power $\dot{N}\hbar\omega$ matching the Unruh thermal flux. Each individual photon carries exactly one quantum of action $\hbar$ and one binary helicity outcome, one bit in the sense of one binary alternative carrying $\ln 2$ nats, and these are paired by the construction independently of a. The Landauer bound then operates as a consistency condition on the ensemble: the minimum energy cost per bit acquired, summed over the rate of acquisition, equals the power required to sustain the acceleration.

3.6 Polarization Bogoliubov Transformation

We now show that the polarization labels survive the Rindler–Unruh mode decomposition, so that the thermal ensemble retains its helicity structure and the anticorrelation of the emitted pair descends unambiguously to the level of individual Rindler microstates.

In the transverse plane, the electromagnetic field in double-null gauge decomposes into two independent scalar-like sectors labelled by helicity $\lambda = \pm 1$. For each sector, the right-wedge field expands as

$$A^i_\lambda(z) = \int_0^\infty \frac{d\omega}{2\pi}\,\frac{1}{\sqrt{2\omega}}\,\epsilon^i_{(\lambda)}\left[b^R_{\omega,\lambda}\,u^R_\omega(z) + b^{R\dagger}_{\omega,\lambda}\,u^{R*}_\omega(z)\right], \tag{21}$$

where $u^R_\omega(z)$ are the right-wedge Rindler modes at Rindler frequency ω, and $b^R_{\omega,\lambda}$, $b^{R\dagger}_{\omega,\lambda}$ are the corresponding annihilation and creation operators satisfying

$$\left[b^R_{\omega,\lambda},\, b^{R\dagger}_{\omega',\lambda'}\right] = \delta(\omega-\omega')\,\delta_{\lambda\lambda'}. \tag{22}$$

An identical expansion holds in the left wedge with $R \to L$ and $\lambda \to -\lambda$, reflecting the anticorrelated helicity assignment. Because each helicity sector is independently a scalar-like sector, the Bogoliubov transformation relating Minkowski and Rindler modes takes the same form in each sector. For helicity λ at Rindler frequency ω, the Unruh modes are

$$c_{\omega,\lambda} = \cosh\theta_\omega\; b^R_{\omega,\lambda} - \sinh\theta_\omega\; b^{L\dagger}_{\omega,-\lambda}, \tag{23}$$

where $\tanh\theta_\omega = e^{-\pi\omega c/a}$ encodes the Unruh distribution at acceleration a. The helicity flip $\lambda \to -\lambda$ on the left-wedge operator is not an independent assumption but a direct consequence of the Bisognano–Wichmann modular conjugation J, which acts as $J\,b^{R\dagger}_{\omega,\lambda}\,J^{-1} = b^{L\dagger}_{-\omega,-\lambda}$ and therefore requires the left-wedge partner of a right-wedge mode of helicity λ to carry helicity $-\lambda$; the PCT symmetry between wedges is thereby reflected in the superposition over both helicity assignments in the Bell state. The transformation (23) is the standard Bogoliubov relation for a massless field [10, 23], now carrying a helicity label throughout. The Minkowski vacuum thermofield double state, in the Rindler basis reads, for each helicity sector,

$$|0\rangle^{(\lambda)}_M = \frac{1}{\cosh\theta_\omega}\sum_{n=0}^\infty \tanh^n\theta_\omega\;|n_{\omega,\lambda}\rangle_R\,|n_{\omega,-\lambda}\rangle_L, \tag{24}$$

and the full vacuum is the product over all ω and both helicities:

$$|0\rangle_M = \prod_{\omega,\lambda} |0\rangle_M^{(\lambda)}. \tag{25}$$

Equation (24) makes explicit that the Minkowski vacuum is a sum over entangled helicity-anticorrelated pairs at each Rindler frequency. The bilocal source specifies a physical emission event: exactly one pair, at fixed ω, with fixed helicity assignment. The PVM projection of Section 3.2 identifies this event with the $n = 1$ term in the vacuum decomposition (24) at that ω and helicity pair. The higher n terms are not suppressed by any weak-coupling assumption; they correspond to different physical events, two-pair emissions, three-pair emissions, and so on, that this source simply does not describe. The selection is prescriptive, not perturbative. The resulting single microstate is

$$|\Psi_\omega\rangle_M = \frac{1}{\sqrt{2}} \left(|1_{\omega,+1}\rangle_R \, |1_{\omega,-1}\rangle_L + |1_{\omega,-1}\rangle_R \, |1_{\omega,+1}\rangle_L \right). \tag{26}$$

$\mathcal{R}$'s and $\mathcal{L}$'s helicity outcomes are undetermined until measurement, at which point they are perfectly anticorrelated by the Bell state structure: each observer acquires one classical bit, and the two bits are complements of one another.

The key structural result is that the helicity label λ is conserved under the Bogoliubov transformation: it enters equation (23) as a spectator index and exits unchanged on the Unruh operators $c_{\omega,\lambda}$. Thermality, when it arises from coarse-graining over the ensemble of microstates (26), is therefore helicity-resolved: each helicity sector thermalises independently, and the anticorrelation between wedges is preserved at every level of the construction.

3.7 Two Geometric Regimes

The microstate (26) admits two complementary geometric descriptions depending on the transverse localisation of the photon mode, corresponding to the two limits of the transverse Kraus family

$$K_{\sigma_\perp} = \int e^{-r^2/2\sigma_\perp^2} \, |\mathbf{x}_\perp\rangle\langle\mathbf{x}_\perp| \; d^2x_\perp, \tag{27}$$

parametrized by $\sigma_\perp$, the direct transverse analogue of the longitudinal Kraus family (14) with the null coordinate u replaced by the transverse radial coordinate $r = |\mathbf{x}_\perp|$. The bilocal source fixes a definite Rindler frequency ω but imposes no constraint on transverse momentum. A mode with zero transverse momentum is uniform across the horizon cross-section; a wave packet of characteristic width $\sigma \sim c/\omega$ is localised within one wavelength of the photon axis. These two limits correspond to geometrically distinct pp-waves at the Rindler horizon, both analysed in Section 5:

- **Delocalized limit** ($\sigma \to \infty$, POVM $\to \mathbf{1}$): the stress tensor is uniform across the horizon, producing a *planar impulsive wave* with longitudinal shift $\Delta v \propto -r^2$, which is a supertranslation in the BMS sense.
- **Localized limit** ($\sigma \to 0$, POVM $\to$ PVM): the stress tensor concentrates near the beam axis, recovering the Aichelburg–Sexl pp-wave with logarithmic shift $\Delta v \propto -\ln r^2$. All BMS supertranslation modes are excited.

Both limits produce the same total horizon area increment ($\Delta A = 8\pi G\hbar\omega$ equation (35) upcoming). The physical reason is that the Raychaudhuri equation integrates the null energy T_{uu} over the entire transverse horizon cross-section: regardless of how that energy is distributed transversely, the integral $\int T_{uu} \, du \, d^2x_\perp = \hbar\omega$ is fixed by energy conservation alone, so the total area increment is insensitive to the transverse profile. The difference between the two limits lies entirely in the spatial distribution of that increment across the horizon and the corresponding BMS mode content excited by each profile.

4 Matter as a Substrate

We propose a language that realizes our microstate as a building block for emergent spacetime. Specifically, we propose that matter is the fundamental substrate of spacetime information processing by reinterpreting quantum mechanics in terms of communication channels. Section 5 will use this language to connect matter action S_{matter} to the Gibbons–Hawking–York boundary term $S_{GHY}|_{\mathcal{H}}$ sample by sample. The following subsections develop this idea:

1. **Matter as a timelike channel** (Section 4.1): We discuss matter action as a measurement of proper time; discreteness lives in the matter sector via $\hbar$. $\mathcal{P}$ and $\mathcal{R}$ are massive observers that accumulate this action.
2. **The horizon as a spacelike channel** (Section 4.2): Spatial geometry emerges from communications between two masses, $\mathcal{P}$ and $\mathcal{R}$, through a horizon, and the Planck area ℓ_P^2 is the conversion factor between the them[5].

In this way, space and time can be measured and quantized as properties of matter and interactions.

Our approach sidesteps a prior question about quantum gravity: *how can one quantize gravity without directly quantizing spacetime or its curvature degrees of freedom?* Common approaches posit Planck-scale cells [24] or discrete spacetime events [25], but Lorentz invariance is recovered only statistically in such constructions rather than at the level of individual configurations [26]. We instead recover discreteness from matter: space and time are properties of matter itself and are defined only up to the resolution set by the action $S/\hbar$, which is Lorentz invariant by construction.

4.1 Matter as a Timelike Channel

A massive particle carries phase along its worldline at a rate set by its rest energy. The Compton frequency $f = mc^2/h$ is the natural clock rate of a mass m: it is the rate at which the phase $e^{iS/\hbar}$ completes one cycle. This is the content of Bremermann's computational limit [27] which describes the maximum clock rate of any physical system with bounded energy.

We associate a discrete event with one zero crossing of the phase $e^{iS/\hbar}$ in the path integral and we use the information-theoretic term *sample* throughout. The sampling resolution is limited by the Heisenberg uncertainty $\Delta E \Delta t \sim \hbar$, which plays the role of the Nyquist limit for the matter channel [28]: it would require additional energy $\Delta E > mc^2$ to probe the sub-Compton phase.

The signal being sampled at the Bremermann rate is the internal quantum state[6] of the particle, its spin, helicity, and other quantum numbers, and the dynamic range is $\log_2$ of the dimension of the internal Hilbert space which motivates:

Definition 4.1. *The channel capacity is the rate at which matter can accumulate and process information, under the assumption that each Bremermann sample accesses the*

[5]The three fundamental constants c, $\hbar$, and G play various roles in the channel picture. The speed of light c sets the causal propagation speed of signals. The reduced Planck constant $\hbar$ sets the resolution of the timelike channel via the Compton frequency $mc^2/\hbar$. The gravitational constant G, through the Schwarzschild radius for instance $r_s = 2Gm/c^2$, sets the maximum energy storable in a spatial region without gravitational collapse. Their combination $\ell_P^2 = \hbar G/c^3$ is the Planck area, the minimal spatial region per independent quantum of action.

[6]For composite systems the internal Hilbert space includes additional degrees of freedom beyond spin and helicity: relative positions and momenta of constituents, vibrational modes, and entanglement between subsystems. How these contribute to the accessible dimension per Bremermann sample, and whether they generate additional spatial structure analogous to the spatial bit of Section 6, are questions the present framework motivates but does not address.

full d-dimensional internal Hilbert space. For a system of mass m with internal Hilbert space dimension d, we define the maximum rate of information flow as

$$C = \underbrace{\frac{mc^2}{h}}_{bandwidth} \times \underbrace{\log_2(d)}_{dynamic\ range} \qquad \textit{(bits per second)}, \tag{28}$$

where the bandwidth is the Bremermann sampling rate and the dynamic range is the information content per sample

Note that both bandwidth and dynamic range are extensive under composition:

$$\frac{mc^2}{\hbar} = \frac{(m_1 + m_2)c^2}{\hbar} = \frac{m_1 c^2}{\hbar} + \frac{m_2 c^2}{\hbar}, \tag{29}$$

$$\log_2(d) = \log_2(d_1 d_2) = \log_2(d_1) + \log_2(d_2), \tag{30}$$

consistent with their interpretation as properties of matter that compose naturally under union.

Whether C is saturated at the event horizon of a black hole, where the Landauer and Bremermann limits are expected to coincide, is left to future work.

Recent experimental demonstrations of the Compton clock [31] ground the matter-as-clock picture in laboratory physics.

4.2 The Horizon as a Spacelike Channel

The horizon enters the channel picture in a double role. As a *channel*, it is the surface through which causal connection is established: the photon crosses it and that crossing is a communication, converting quantum entanglement between the wedges into a classical record in matter-memory. As a *barrier*, it is the surface at which this conversion makes the opposite wedge causally irrelevant to the receiving observer's future evolution. The disconnection between wedges is therefore not a pre-existing feature of the geometry that the photon happens to cross; it is a consequence of the communication itself. Before the crossing $\mathcal{R}$ and $\mathcal{L}$ share a global pure state; after it, $\mathcal{L}$ is inaccessible to $\mathcal{R}$ precisely because the entanglement has been consumed in producing the classical record.

The photon crossing the horizon is a transfer of information and action between the timelike matter channels of $\mathcal{R}$ and $\mathcal{L}$, mediated by the spacelike horizon channel. To that end we define:

Definition 4.2 (Communication). *A communication is a horizon-crossing particle*[7] *that transfers action and information between matter-memories.*

In the explicit bilocal construction of Section 3, this takes the following specific form:

(i) a massless quantum of Rindler frequency ω and helicity $\lambda = \pm 1$, selected by the PVM projection of Section 3.2, transferring one quantum of action $\hbar$ across $\mathcal{H}$;

(ii) the receiving timelike channel incrementing its Bremermann sampling rate by $\Delta f = \omega/2\pi$.

(iii) the helicity outcome λ recorded as one classical bit in the receiving channel's matter-memory; and

(iv) the horizon area incrementing by $\Delta A = 8\pi G\hbar\omega$.

Property (iv) follows from the Raychaudhuri equation applied to the null congruence at $\mathcal{H}$ and is derived in Section 5; it is listed here to make the geometric content of the communication explicit. Whether properties (i)–(iii) generalize beyond the photon case, to

[7]For our microstate this is a photon, however, it could be more general.

massive carriers, higher-spin fields, or POVM rather than PVM projections, is a question the construction motivates but does not answer.

Each communication has exactly two irreducible constituents: one quantum of action $\hbar$ written into matter-memory, incrementing the Bremermann sampling rate, and one bit of classical information, the helicity outcome. These are not independent: they are the energy $\hbar\omega$ and angular momentum $\pm\hbar$ of a single photon, two properties of the same physical object. With this definition in hand, the equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ of Section 5 becomes a counting statement: the GHY boundary term sums $\hbar\omega$ over communications because that is what the term geometrically records, one contribution per event.

4.3 Relation to Existing Literature

The matter-substrate language of this section makes contact with two independent programs in quantum foundations that approach emergent time from complementary directions.

The Page–Wootters mechanism [32, 33] derives time evolution relationally: a global static state $|\Psi\rangle \in \mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{sys}}$ satisfying a Wheeler–DeWitt-like constraint yields emergent dynamics when conditioned on clock eigenstates $|t\rangle$. The mechanism is exact but leaves the physical identity of the clock system unspecified. The present framework supplies that identity: the matter substrate is the clock, with phase $e^{iS/\hbar}$ advancing at the Bremermann rate mc^2/h. The standard mechanism assumes a continuous clock time t; the present construction replaces this with a discrete sequence of communication units, each acting as a PVM conditioning step that advances the Bremermann count by one quantum of action. Proper time is relational in exactly the Page–Wootters sense, but the clock states are physical rather than abstract.

The modular flow perspective of Connes and Rovelli [34] approaches the same problem algebraically. In a generally covariant quantum theory the Hamiltonian vanishes on physical states and no preferred time exists; Connes and Rovelli show that a state ω on a von Neumann algebra $\mathcal{M}$ nonetheless generates a canonical one-parameter automorphism group, the modular flow σ_t^ω, via Tomita–Takesaki theory. This flow satisfies the KMS condition and is thermodynamically indistinguishable from thermal time evolution. The Bisognano–Wichmann theorem [5], already central to the PCT structure of Section 3, identifies this modular flow concretely with Rindler time evolution for the vacuum restricted to one wedge. The connection to the present framework is therefore already implicit in the construction: the horizon $\mathcal{H}$ is precisely the algebraic cutoff that Connes and Rovelli require, restricting the observable algebra to the right wedge and generating the modular flow as a consequence. What the present framework adds is a discrete substrate beneath that flow: the continuous modular parameter is the thermodynamic limit of discrete Bremermann samples.

The two programs are thus not independent supports for emergent time but complementary descriptions of the same structure at different levels: Page–Wootters gives the kinematic conditioning on individual clock states, Connes–Rovelli gives the thermodynamic flow that emerges when those states are coarse-grained across a horizon. The matter substrate and the communication units are the microscopic objects that both programs require but neither specifies.

Verlinde's entropic gravity [2] operates at the thermodynamic level of the same horizon structure; the present framework supplies the microstate layer beneath it, with each communication realising one entropy increment of the kind Verlinde's screens accumulate in bulk. The Wigner's friend problem [35] and the Frauchiger–Renner theorem [36] identify the multi-observer problem as formally sharp; the vocabulary introduced here, spatial bits, PVM projections, communication units, matter-memory, is the natural language in which that problem must be posed when the observers are causally separated by a horizon.

5 pp-Wave Geometry and Action

We work in the null coordinates u, v defined in Section 3, with $\mathcal{P}$ at $(u, v) = (0, 0)$, $\mathcal{R}$ at $(0, 2)$, and the photon propagating along $u = 0$ and intersecting the past right Rindler horizon at $(0, 1)$; the left wedge is symmetric.

5.1 The AS pp-Wave and Its Effects

Test particles at fixed transverse positions (x, y) experience two effects as the shock front crosses $u = 0$:

1. **Longitudinal shift.** The v coordinate is displaced by

$$\Delta v \propto -\ln r^2, \qquad r^2 = x^2 + y^2. \tag{31}$$

   This shift is lightlike: no proper time elapses for a particle crossing the shock, but it is displaced along the null direction. Near the axis ($r \to 0$) the shift diverges, encoding the concentration of energy along the photon's trajectory.

2. **Transverse focusing.** The gradient of the longitudinal shift produces a transverse velocity

$$\Delta \mathbf{v}_\perp \propto \nabla \ln r \propto \frac{1}{r}, \tag{32}$$

   directed toward the axis. Test particles are pulled toward the centre of the wave, with the kick strength growing as $1/r$ toward the axis.

5.2 The Longitudinal Shift Makes Space

The longitudinal shift deserves closer attention, because it carries a meaning beyond a mere coordinate displacement. The photon emitted by $\mathcal{P}$ propagates along the null direction; as it crosses the Rindler horizon, the AS shift displaces the future light cone of every test particle it passes. For the photon traveling into the right wedge, the shift pushes the right wedge's geometry in the $+v$ direction; for the entangled partner traveling into the left wedge, the shift pushes the left wedge in the $+u$ direction. The two wedges are displaced *away from each other* along the longitudinal null direction. See Figure 5.

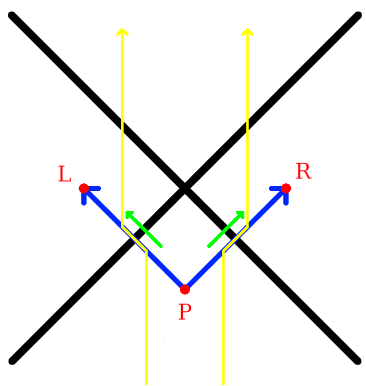

Figure 5: The green arrows show the longitudinal shift that occurs where the photons cross the horizon. Test particles, shown in yellow, are pushed outwards away from $z = 0$ due to the creation of space along the null lines originating from $\mathcal{P}$.

At the intersection of the shock front $u = 0$ with the past right Rindler horizon at $(u, v) = (0, 1)$, the transverse (x, y) plane is identified with the horizon area elements $dA = dx\, dy$. The longitudinal shift $\Delta v \propto -\ln r^2$ displaces the horizon outward in the

v direction, pushing the right wedge geometry away from the horizon. The entangled partner photon produces a symmetric shift in the u direction at the left Rindler horizon, pushing the left wedge away from its horizon. The two wedges are therefore displaced away from their respective horizons in opposite null directions.

5.3 Transverse Focusing Increments Horizon Area

The area increment Δ is independent of the transverse localization of the photon. In the delocalized limit the stress tensor is uniform across the horizon and produces a planar shift $\Delta v \propto -r^2$; in the localized limit it concentrates near the beam axis and produces the AS logarithmic shift $\Delta v \propto -\ln r^2$. Both are supertranslations in the BMS sense and both give the same area increment, as the Raychaudhuri calculation below shows. The one-parameter family of Gaussian transverse profiles interpolating between these two limits, parametrized by width σ, is the geometric counterpart of the POVM-to-PVM transition of Section 3.2: the spatial distribution of the area increment sharpens from uniform to concentrated as $\sigma \to 0$, while the total ΔA remains fixed by energy conservation throughout.

We derive the area increment using the Raychaudhuri equation on the null congruence generating the Rindler horizon, following Jacobson [1] and consistent with the generalized second law for Rindler horizons proved by Wall [37], but applied here to a single microstate rather than a thermodynamic or ensemble average.

For a null congruence with generators k^μ and expansion θ, the Raychaudhuri equation gives

$$\frac{d\theta}{du} = -\tfrac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - 8\pi G\, T_{\mu\nu}k^\mu k^\nu. \tag{33}$$

Integrating over the horizon patch in the linearised regime and using $k^\mu = \delta^\mu_u$, the area change is

$$\Delta A = 8\pi G \int T_{uu}\, du\, d^2x_\perp. \tag{34}$$

The present construction provides a natural regularization of the strict AS point-source limit: the photon is a normalizable mode of the electromagnetic field in double-null gauge, with a smooth transverse profile set by the wavelength $\sim c/\omega$. The electromagnetic stress-energy T_{uu} is well-defined throughout, integrates to $\hbar\omega$ over the transverse plane by energy conservation, and recovers the strict AS limit as $\omega \to \infty$ with the logarithmic divergence regulated at the scale c/ω.

Substituting into equation (34) gives

$$\boxed{\Delta A = 8\pi G\,\hbar\omega = 8\pi G\,\Delta E.} \tag{35}$$

The horizon area grows by one discrete, calculable increment per photon absorbed, microstate by microstate, without reference to bulk thermodynamic averages. This is property (iv) of the communication definition of Section 4.2 verified geometrically: the area increment $\Delta A = 8\pi G\hbar\omega$ is the geometric expression of one quantum of action $\hbar$ deposited into matter-memory by one measurement event, with the Planck area ℓ_P^2 as the conversion factor between the two descriptions.

Here the null generators of the horizon are taken to be affinely parametrised as $k^\mu = (\partial/\partial u)^\mu$ in double-null coordinates, so that $\int T_{uu}\, du\, d^2x_\perp = \hbar\omega$ is the total null energy of the photon. This differs from the Killing-normalised convention by a factor of the surface gravity $\kappa = a$; the two are related by $(\partial/\partial u)^\mu = a(\partial/\partial\eta)^\mu$ at the horizon, consistent with the standard result $\Delta A = 8\pi GE/\kappa$ when $E = \hbar\omega$ is the Killing energy.

The longitudinal shift $\Delta v \propto -\ln r^2$ produced by each pp-wave at the Rindler horizon is a supertranslation in the sense of Bondi–Metzner–Sachs (BMS) symmetry. Hawking, Perry, and Strominger [38] showed explicitly that a black hole is supertranslated by the passage of an asymmetric shockwave, and the AS longitudinal shift is precisely the linearised supertranslation field on the horizon. Carlip [39] demonstrated that the near-horizon diffeomorphisms of a generic black hole are enhanced to a BMS_3 algebra whose central charge determines the Bekenstein–Hawking entropy, but his program identifies the symmetry without specifying the physical process that excites individual modes. The bilocal construction supplies that process: each measurement event produces one AS pp-wave and deposits one supertranslation at the Rindler horizon. The shockwave S-matrix program of 't Hooft [40] established that the longitudinal AS shift is the fundamental dynamical ingredient for horizon information exchange at the level of the full S-matrix; the present construction realises the same shift at the level of a single microstate.

5.4 Action and the Gravitational Boundary

The equality derived below is an internal consistency check on the microstate identification, not a novel result claimed in isolation. The thermodynamic interpretation of the GHY boundary term on null surfaces is established by Chakraborty and Padmanabhan [41]; the broader program of Padmanabhan [42] identifies gravitational boundary terms as the fundamental degrees of freedom at the ensemble level. What the present construction contributes is not the identity $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ but its realisation microstate by microstate: if individual quantum measurement events genuinely are the microstates underlying each area increment, the geometric count and the informational count must agree event by event. The calculation below verifies that they do, closing the consistency loop.

Each absorption event transfers null energy $\hbar\omega_i$ across the horizon. The matter action at the horizon is defined as the null-surface integral of the stress-energy, evaluated over the horizon generators:

$$S_{\text{matter}} \;=\; \sum_i \int T^{(i)}_{uu}\, du\, d^2x_\perp \;=\; \sum_i \hbar\omega_i, \tag{36}$$

where the second equality follows from energy conservation applied to each photon. This is the null-surface analog of the worldline action $\int E\, d\tau$: on a null surface the affine parameter u plays the role that proper time plays on a timelike worldline, and the integral $\int T_{uu}\, du$ plays the role of $\int E\, d\tau$.

The two are not the same integral. After the absorption event, the detector's worldline action continues to accumulate as $\hbar\omega_i\, \Delta\tau$ for all subsequent proper time $\Delta\tau$, encoding the Bremermann memory of the event. The equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ holds for the null-surface version of the matter action, evaluated at the moment of horizon crossing; it is a statement about the communication itself, not about the subsequent worldline evolution. The Rindler proper time η and the affine null parameter u are related by $u \propto e^{-a\eta}$, and it is this exponential relation, not a simple proportionality, that is responsible for the Unruh temperature and that distinguishes the null-surface action from the worldline action. This is the matter action counted sample by sample: one quantum of action per event, one event per emitted pair.

The same events have a geometric description. By eq. (35), each absorption sample contributes a horizon area increment $\Delta A_i = 8\pi G\hbar\omega_i$, so the total area increment summed over all events is

$$\sum_i \Delta A_i = 8\pi G \sum_i \hbar\omega_i = 8\pi G\, S_{\text{matter}}. \tag{37}$$

This geometric sum has a precise identification in the gravitational action. On a stationary Rindler horizon the unperturbed expansion of the null generators vanishes, $\theta = 0$, so the Gibbons–Hawking–York boundary term [43] is sourced entirely by the AS pp-wave perturbations:

$$S_{GHY}\big|_{\mathcal{H}} = \frac{1}{8\pi G}\int_{\mathcal{H}} \theta\, d^2x_\perp\, du = \frac{1}{8\pi G}\sum_i \Delta A_i = \frac{1}{8\pi G}\cdot 8\pi G\sum_i \hbar\omega_i = \sum_i \hbar\omega_i = S_{\text{matter}}. \tag{38}$$

Chakraborty and Padmanabhan [41] showed that the gravitational boundary term on null surfaces has a thermodynamic interpretation as a heat density Ts of the null surface, establishing $S_{GHY}|_{\mathcal{H}}$ = heat content at the level of thermodynamic averages. The present result realises the same identity at the level of individual microstates: each contribution to S_{GHY} corresponds to one quantum of action deposited into matter-memory by one measurement event, precisely the communications of Section 4.2. The Raychaudhuri equation and the communication definition count the same events in two languages, and they agree.

This result is consistent with and extends programs of Jacobson [1] and Padmanabhan [42], which argue that the gravitational boundary term rather than the bulk Einstein–Hilbert action is the fundamental gravitational degree of freedom. Those programs establish this at the level of thermodynamic averages. The present construction realizes the same identification at the level of individual microstates, with each boundary contribution sourced by a specific quantum measurement event rather than a bulk average. The standard result emerges from summing over the ensemble; the microstate picture supplies what the thermodynamic programs leave open: the physical process underlying each boundary contribution.

6 Entanglement as Spatial Information

Van Raamsdonk [17] showed that the entanglement entropy between two halves of a thermofield double state controls the geometry connecting them: reducing entanglement elongates the ER bridge until the geometry disconnects entirely. The vacuum decomposition (24) is precisely this thermofield double state, now in flat Minkowski space with helicity labels carried through. The bilocal construction realises Van Raamsdonk's argument event by event: in the delocalized limit the full thermofield double produces a global BMS supertranslation displacing both wedges uniformly, while in the localized limit each individual measurement event produces a single AS longitudinal shift concentrated near the photon trajectory, displacing the two wedges apart in opposite null directions and creating new coordinate space between them; in the AdS setting this is the ER=EPR correspondence [44].

In the present flat-space construction the ER bridge is replaced by the AS longitudinal shift: each measurement event displaces the two wedges apart in opposite null directions, creating the coordinate space between them that the ER bridge represents in the AdS setting. The entanglement is not a property of the geometry; it is the *cause* of it, event by event.

The bilocal source prepares an anticorrelated Bell state (26). Both $\mathcal{R}$ and $\mathcal{L}$ measure in the same basis and obtain definite outcomes, which we label H_+ or H_-, perfectly anticorrelated by the Bell state structure independently of which basis is chosen. The question

"who observed H_+?"

has a binary answer: $\mathcal{L}$ (left wedge) or $\mathcal{R}$ (right wedge). This is a *spatial bit*: a bit that now encodes not a polarization state but a location in space.

7 Conclusion

The standard picture of spacetime puts geometry first: matter lives in space, moves through time, and gravity tells it how. This paper inverts that priority. Matter is the substrate: proper time is action accumulated along worldlines, space is the record of communication units at horizons, and the Planck area ℓ_P^2 is the conversion factor between the two descriptions. Geometry is not the stage on which physics happens; it is the ledger in which matter's interactions are recorded.

The horizon is where this inversion becomes sharp. As a causal surface it is ontic: signals cannot cross from $\mathcal{L}$ to $\mathcal{R}$. As an epistemic boundary it is the surface at which a single observer's description becomes irreducibly incomplete, not because the physics is hidden but because the information is held by someone else. The Bekenstein–Hawking entropy is the measure of that gap, and the Planck area is its conversion factor into geometry. Thermality is not a primitive feature of the vacuum; it is what remains when the positions of $\mathcal{P}$ and $\mathcal{R}$ are unspecified and the other observer's record is inaccessible. Specify both, and the thermal ensemble resolves into individual non-thermal microstates, each a single measurement event, each contributing one Planck-area patch to the horizon.

The deepest consequence is the spatial bit. The same event that records *what* was observed also determines *where*: the two wedges carry opposite helicities by construction, so the polarization outcome and the location are the same classical fact written into matter-memory. Four causally separated observers, $\mathcal{P}$, $\mathcal{R}$, $\mathcal{L}$, $\mathcal{F}$ (a future version of $\mathcal{P}$), each hold partial records of the same physical process, and their consistency is enforced not by an objective wavefunction but by common causal ancestry. This is the Wigner's friend scenario instantiated at a horizon, and it is not resolved here. What the construction shows is that the vocabulary needed to pose that problem precisely, spatial bits, communications, matter-memory, PVM projections, the Planck area as conversion factor, is the same vocabulary in which the answer, when it comes, will have to be written. Space, in this construction, is a consequence of helicity. We leave it there.

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9 Appendix: The Tick-Tock Mechanism

The communication interactions from Section 4.2 have a natural analogue along a timelike worldline. Penrose's zig-zag construction [45] represents a massive fermion as two massless Weyl spinors of opposite helicity, coupled at each vertex by the Higgs field and alternating in a zig-zag pattern; we term this the tick-tock mechanism to emphasise the alternating rhythm of communications rather than the geometry of the path. The parallel with the bilocal construction is structural: the Higgs field plays the role of $\mathcal{P}$, the two Weyl spinors play the role of the two photons propagating to $\mathcal{R}$ and $\mathcal{L}$, and each vertex is a communication unit depositing one quantum of action and producing one sample of proper time rather than one increment of horizon area. We present this as a suggestive analogy rather than a derived result.

The anticorrelated helicity assignment in both constructions has a common algebraic origin. The Bisognano–Wichmann modular conjugation J acts on right-wedge photon

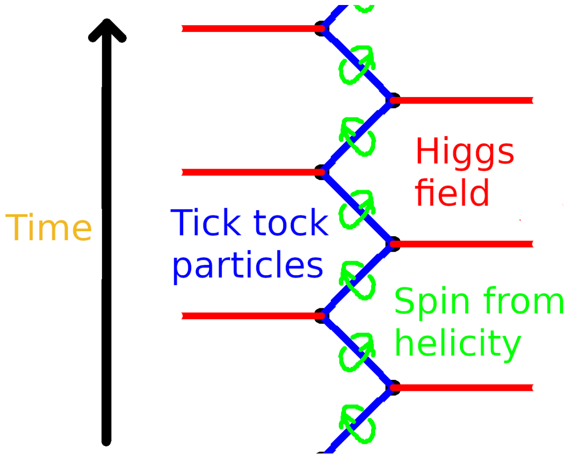

Figure 6: The tick-tock mechanism. A massive particle (blue) alternates between right- and left-helicity massless segments, coupled at each vertex by the Higgs field (red). Spin emerges from the alternating helicity flip at each vertex (green loops), visible as the two $\mathfrak{su}(2)$ factors of $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$ cycling between communications. Each vertex is a communication event: one quantum of action deposited, one helicity outcome recorded, one sample of proper time produced.

creation operators as

$$J\, a^{R\dagger}_{k,\lambda}\, J^{-1} = a^{L\dagger}_{-k,-\lambda}, \tag{39}$$

flipping helicity and reversing the spacelike separation direction $R \leftrightarrow L$. The PCT operator Θ acts on a right-handed Weyl spinor as

$$\Theta\, \psi_R(t, \mathbf{x})\, \Theta^{-1} \propto \psi_L(-t, -\mathbf{x}), \tag{40}$$

flipping helicity and reversing the timelike separation direction $t \to -t$. In both cases the operation is the same at the representation-theoretic level: exchange of the two $\mathfrak{su}(2)$ factors of $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$. The operators J and Θ act on different Hilbert spaces; what they share is that both select anticorrelated helicities as the unique PCT-invariant configuration.

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